Md Shahanur Islam Shagor
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I am writing to apply for the Systems Software Engineer (Rust, ML Inference) role at ai-coustics. Your work on building the reliability layer for Voice AI, combining real-time audio processing with production-grade SDKs, is exactly the kind of infrastructure challenge I want to be part of.

Through hands-on project development, I have built and optimized real-time ML inference systems that run on constrained hardware. One example is a wireless vision-aid system that uses YOLOv8n for object detection, with a custom UDP streaming protocol to maintain ultra-low latency. This required careful memory management, performance profiling, and ensuring deterministic behavior under strict timing constraints. I also developed a privacy-preserving blockchain framework for autonomous vehicle networks, where I worked on secure communication protocols and cryptographic data integrity using C++ and Python.

What stands out to me about ai-coustics is the focus on making neural audio models run fast and reliably outside of research environments. I have experience bridging that gap. In my work on the WVAB system, I handled model conversion, local inference optimization, and cross-platform deployment on devices like Raspberry Pi and ESP32-CAM. I am comfortable reasoning about trade-offs between performance, safety, and portability, and I enjoy building tooling that makes ML models easier to test, benchmark, and deploy.

In addition to my technical experience, I work effectively across multidisciplinary teams and communicate clearly with both technical and non-technical stakeholders. I am fluent in Bengali and English, have working knowledge of Russian and Hindi, and enjoy collaborating in international and cross-cultural environments.

I have attached my CV for your review and would welcome the opportunity to discuss how I can contribute to the Systems team at ai-coustics.

Sincerely,

Md Shahanur Islam Shagor
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